Andrew Pols

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Education

University of Toronto **Sep. 2024 – May 2028**
- HBSc., **Computer Science and Statistics** **GPA: 4.00 / 4.00**
- Key coursework: Software Design; Computer Organization; Data Science; Software Tools and Systems Programming

Experience

Software Developer Intern **May 2026 – May 2027**
IBM Canada Ltd.
- Shipped a dependency graph to all Planning Analytics Workspace clients, built in React and TypeScript. It lets them interactively trace and filter relationships in their financial models of thousands of objects while staying responsive.
- Cut the initial load from **30s to 3s** on a 2,000-object model, with **50 requests down to 1** and each later object under **500ms**, using lazy-loading, request aggregation, and caching. This turns a manual trace from hours down to minutes.
- Turned a manual workflow into requirements for graph layout, filtering, and interactions with design, product management, QA, and engineering in Agile sprints, and owned the whole feature from design to production release.
- Feature shown to IBM leadership and at TM1 Anwendertag and TechXChange conferences, with excellent feedback.
- Building a text widget for the modernized dashboarding experience that keeps text readable across theme and widget color changes, and integrating LLM-based localization so teammates in other locales see saved text in their language.

Projects

ScholarSearch | *Source Code* **Python | FastAPI | PostgreSQL | ChromaDB | Redis | ReactJS**
- Built a hybrid BM25 and semantic search engine over **1,000,000 DBLP papers**, serving cold queries in a **334ms** median (p99 under 600ms) and cached repeats in **1.2ms**, plus a React frontend with interactive D3.js graph views.
- Embedded all 1M papers at 313 papers/sec and traced the 1M-scale slowdown to the 300ms in-memory BM25 scan.

FindMySound | *Source Code* **Python | Django | SQLite | Scikit-Learn | ReactJS**
- Built a web app that recommends music from a user's library using Spotify's Web API, OAuth 2.0, and cosine similarity on BERT feature vectors, reaching **95% satisfaction** across 5,000 recommended songs (50 users, 100 songs each).
- Cached every song's audio analysis (about 2s per song) so scoring a candidate takes **0.2ms**. Built the Django backend with a custom user model, JWT authentication, SQLite database, plus REST API endpoints for the React frontend.

SudokuCV | *Source Code* **Java | OpenCV | Tesseract OCR**
- Built a Java app that solves Sudoku from an image or webcam feed using OpenCV, Tesseract OCR, and recursive backtracking, finding the grid in **about 2ms per frame** and solving **27 of 27** test puzzles in a median of 0.5ms each.
- Raised exact grid reads from **7% to 92%** and digit accuracy from 89% to **99.96%** on 789 sampled frames of phone footage of 3 printed Sudoku puzzles after fixing a bug in Hough-based grid-line removal that erased stacked digits.

Multiplayer Pokémon Battle Server | *Source Code* **C | TCP Sockets | UNIX | ncurses**
- Built a multiplayer Pokémon battle server in C that holds **1,000 concurrent TCP clients** in a single-threaded select() loop at **about 54k messages/sec inbound**, with p99 latency near 25ms on loopback and zero connection failures.
- Pairs players into 1v1 matches with a lobby, rematches, forfeits, live chat, and an ncurses CLI interface on the client.
- Defined a binary packet protocol with a 5-byte header (a command byte and length in network byte order), buffering reads so that fragmented TCP packets parse correctly and the other client is notified when an opponent disconnects.

Technical Skills

- **Programming Languages**: Python, TypeScript, JavaScript, Java, C, Go, SQL, HTML5, CSS
- **Frameworks & Libraries**: React, Node.js, FastAPI, Django REST, Flask, Next.js, Scikit-learn, OpenCV
- **Databases & Cloud**: SQL (PostgreSQL, MySQL), NoSQL (MongoDB, Redis), ChromaDB; AWS, Azure, Docker
- **Developer Tools**: Git (version control), REST APIs, IBM Bob, Jira, UNIX, Nginx, Valgrind, AddressSanitizer